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Measurement based Characterization of Electromagnetic Noise for Industrial Internet of Things

Yuan Ze[a], Liu Liu[a, b, c, *], Zhang Kun[a], Zhang Jianhua[b]

[a]*School of Electronic and Information Engineering, Beijing Jiaotong University, Beijing 100044 , China*
[b] *Key Laboratory of Universal Wireless Communication, Ministry of Education, Beijing University of Posts and Telecommunications, Beijing 100876, China*
[c]*National Mobile Communications Research Laboratory, Southeast University, Nanjing 210096, China.*

Abstract

The Industrial Internet of Things (IIOT) can help the manufacturing enterprises improve the production efficiency, reduce costs and achieve intelligent factory production. In this harsh scenario, however, the electromagnetic noise performs a severe impact on IIOT which uses low-power wireless sensor devices. In this article, we aim to characterize electromagnetic noise for IIoT systems based on the realistic measurement data collected in an automobile factory. The measurements were taken in time and frequency domain into account. By using the frequency-domain measurement data, we can extract the frequency occupation and power amplitude information of electromagnetic noise in 300MHz~3GHz band. By using the time-domain measurement data, the amplitude probability distribution (APD) could be obtained of electromagnetic noise at 315MHz, 779MHz and 916MHz bands. Then, an improved bell shaped spectrum is used to characterize the frequency spectrum of wide band electromagnetic noise.

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Keywords: Industrial Internet of Things; Electromagnetic Noise; Amplitude Probability Distribution

*Liu Liu. Tel.: 13161356596.
E-mail address: liuliu@bjtu.edu.cn

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1. Introduction

The Industrial Internet of Things, a typical use case of mMTC (massive Machine Type of Communication) scene in the fifth generation (5G), attracts widely attentions from both academia and industry [1][2]. However, in such harsh industrial environments, the unwanted electromagnetic noise is relatively high due to the frequency converters, welding equipments, ignition systems, etc. [3][4].

Previous scholars have studied the electromagnetic noise in some industrial environments. In [5], researchers used Markov process to model the electromagnetic noise in factories and compared it with additive white Gaussian noise. The results show that there were differences between them. In [6], scholars measured the electromagnetic noise caused by car ignition, manufacturing car electronic components and shoe production equipment at 2.4GHz.

The remainder of this article is organized as follows. We describe the measurement system and environment. We provide the measurement and modeling results of the time-domain and the frequency-domain. Finally, we discuss the conclusion.

2. Measurement methods and statistical parameters

Our measurements were performed in three locations: near the manual spot welder, near the welder robot and in the office area. We consider two measurement methods: the frequency-domain measurement and the time-domain measurement. A log-periodic antenna and a frequency spectrum analyzer are employed in the measurement. Fig. 1 shows the actual scene of the measurement.

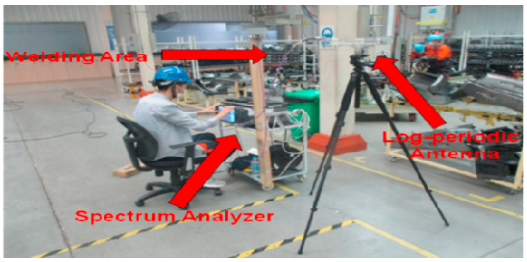

Fig. 1. Actual scene of the measurement

2.1. Frequency-domain Measurement Method

The occupied BW mode of the spectrum analyzer is used in frequency-domain measurement method. We obtain the spectrum occupancy from 300MHz to 3GHz in three measurement locations. All parameters of frequency-domain measurements are listed in Table 1.

Table 1. Parameters of frequency-domain measurements

Parameters	Values
Frequency Band	300MHz~3GHz
Mode	Occupied BW
Span	50MHz
RBW	10KHz
VBW	100KHz
Attenuation	2dB
Preamp	Open
Points	20001

2.2. Time-domain Measurement Method

The data collecting mode of the spectrum analyzer is used in time-domain measurements. The parameters of time-domain measurements are shown in Table 2.

Table 2. Parameters of time-domain measurements

Parameters	Values
Frequency	315MHz、779MHz、916MHz
Bandwidth	200KHz
Sampling Rate	2MHz
Measurement Duration	134.2 s
Mode	Data Collecting

2.3. Statistical Parameters

APD is defined as the fraction of time that the noise amplitude is greater than a threshold level and is expressed as [7]

$$APD(L_i) = P(\lambda > L_i) = \frac{1}{T}\sum_{k=1}^{N} t_k \tag{1}$$

Where λ is the noise amplitude, L_i is the threshold level, T is the measurement duration, t_k is the pulse duration exceeding the threshold level.

APD can also be expressed as

$$APD(L_i) = P(\lambda > L_i) = 1 - CDF(L_i) \tag{2}$$

Where $CDF(L_i)$ is the cumulative distribution function of the noise amplitude.

3. Characterization Using the Measurement data

3.1. Frequency-domain Measurement Results

According to the measurement results, we can conclude that the noise frequency distributions of three measured positions are nearly the same. The noise power in the welding workshop is higher than that in the office area. The IIOT is mainly for the welder robot, thus we will discuss the measurement results of the welder robot in the following section.

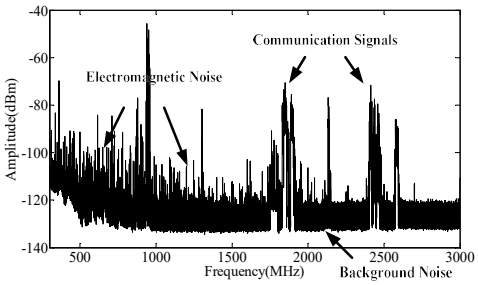

Fig. 2. The spectrum of welder robot in vertical polarization

In Fig. 2, we can find electromagnetic noise, background noise and operating mobile communication signals, such as 2G(GSM) and 4G(LTE) signals. The frequency spectrum occupancy of the communication signals is listed in Table 3. We can see that there are radio and TV signals below 800MHz, and the frequency bands around 900MHz and 1800MHz are operating mobile communication signals.

Table 3. The frequency spectrum occupancy of communication signals

Frequency(MHz)	Signal type	Frequency(MHz)	Signal type
478-486		825-835	CDMA UL
518-526		870-880	CDMA DL
526-534		890-915	GSM800 UL
614-622		935-960	GSM800 DL
654-662		1710-1755	GSM1800 UL
662-670	Radio and TV Signals	1755-1785	FDD-LTE UL
670-678		1805-1840	GSM1800 DL
702-710		1840-1875	FDD-LTE DL
718-726		1885-1915	TD-LTE
758-766		2130-2145	WCDMA DL
766-774		2401-2481	WIFI
790-798		2575-2595	TD-LTE

As can be seen from Fig. 2 and Table 3, the signals having broadband and high power are the operating mobile communication signals. The electromagnetic noise is mainly below 1GHz, and has small power level. We can also conclude that the power of electromagnetic noise is decrease with the increase of the frequency.

3.2. Modeling of Electromagnetic Noise Spectrum

The frequency spectrum of the wideband electromagnetic noise is not a sharp pulse in frequency domain (as shown in Fig. 3), and it can be characterized as a bell shaped spectrum which is defined in 802.11n as [8]

$$S(f) = \frac{1}{1 + A\left(\frac{f}{f_d}\right)^2}, \quad |f| \le f_{\max} \tag{3}$$

Where A and f_d have a fixed value. In order to model the spectrum of electromagnetic noise, we improve the expression of bell shaped spectrum as

$$S(f) = \frac{P_{\max}}{1 + K\left(f - f_0\right)^2}, \quad |f - f_0| \le f_{\max} \tag{4}$$

Where $p_{\max}$ is the maximum power, f_0 is the center frequency of the electromagnetic noise, $2f_{\max}$ is the bandwidth of the electromagnetic noise, K is a constant to adjust the range of the spectrum amplitude. Results of measured noise and simulation noise at 400.5MHz are shown in Fig. 3. The bandwidth of the measured noise is 32.5KHz.

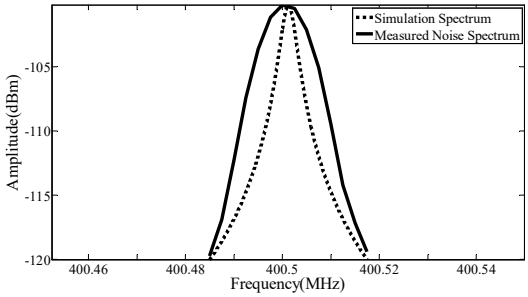

Fig. 3. Spectrum of measured noise and simulation noise

The solid line is the measured noise spectrum, and the dashed line represents the simulation spectrum. Where $K = 3.6\times10^{-7}$, $f_{\max} = 16.25\text{KHz}$, $f_0 = 400.5\text{MHz}$, $P_{\max} = \text{-}100.19\text{dBm}$. The measured noise spectrum is similar to the simulation spectrum according to Fig. 3. So the improved bell shaped spectrum can be used to model the spectrum of electromagnetic noise.

3.3. Time-domain Measurement Results

APD values of the measured noise are shown in Fig. 4. The solid line is the APD of the receiver's thermal noise which is measured without the antenna. Other curves are the APD values extracted in our measurements.

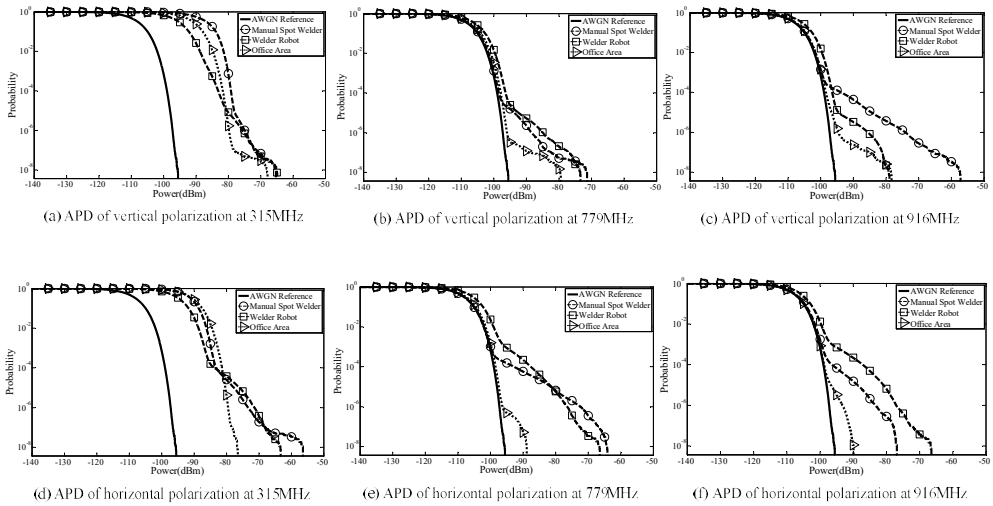

Fig. 4. APD Results

According to Fig. 4, we can find that the APD results measured using the horizontal and vertical polarization is approximately the same. The tails (at the level of 10^{-8}) of the APD shown in Fig. 4 indicate the maximum power levels measured in the office area is less than those measured in the manual spot welder and the welder robot since the office area is less influenced by electromagnetic noise. Compared to the APD results measured in factory infrastructures from [9] [10], we can find that the trend of the APD curves is similar. However, in our results, the noise power is different because of the difference in measurement frequency and bandwidth.

4. Conclusions

In this article, we describe the measurement system and the environment. The measurement results of electromagnetic noise are presented. Frequency-domain measurements obtain the spectrum occupancy of the factory wireless environment in 300MHz~3GHz bands. We present APD measurements from three measurement locations at 315MHz, 433MHz and 916MHz in time-domain measurements. According to the results of measurements, we can conclude that an improved bell shaped spectrum can be used to model the spectrum of electromagnetic noise.

Acknowledgment

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